

C# Programming Quiz Questions

Character Processing and String Manipulation

1. Which of the following is NOT a reason for processing individual characters in a string?

- A) Password validation
- B) Converting strings to integers
- C) Counting specific types of characters
- D) Formatting telephone numbers

Answer: B

2. In C#, what data type is used to store a single character?

- A) string
- B) char
- C) character
- D) single

Answer: B

3. What is the correct way to declare and initialize a char variable with the letter 'a'?

- A) `char letter = "a";`
- B) `char letter = 'a';`
- C) `char letter = a;`
- D) `char letter = (char)"a";`

Answer: B

4. Why can't you assign a string to a char variable, even if the string contains only one character?

- A) Because strings are immutable
- B) Because char and string are incompatible types

- C) Because char can only hold numeric values
- D) Because strings are stored in the heap

Answer: B

5. How do you convert a char variable to a string?

- A) `char.ToString()`
- B) `Convert.ToString(char)`
- C) `(string)char`
- D) Both A and B

Answer: D

6. Which method can convert a single-character string to a char?

- A) `char.Parse()`
- B) `Convert.ToChar()`
- C) Both A and B
- D) `char.ToChar()`

Answer: C

7. How do you access the first character in a string named "name"?

- A) `name[1]`
- B) `name(0)`
- C) `name[0]`
- D) `name.First()`

Answer: C

8. What does the following code return?

```
string str = "Hello";  
char ch = str[10];
```

- A) 'o'

- B) Throws an `IndexOutOfRangeException`
- C) `'\0'`
- D) `'H'`

Answer: B

9. Which loop can be used to iterate through all characters in a string without an index?

- A) `for`
- B) `while`
- C) `do-while`
- D) `foreach`

Answer: D

10. What is true about accessing characters in a string using subscripts in C#?

- A) You can modify characters directly
- B) It provides read-only access
- C) It works only with strings longer than 10 characters
- D) It is slower than using `Substring()`

Answer: B

Character Testing Methods

11. Which method checks if a character is a digit?

- A) `char.IsNumber()`
- B) `char.IsDigit()`
- C) `char.IsNumeric()`
- D) `char.IsInteger()`

Answer: B

12. What does `char.IsLetter('5')` return?

- A) `True`

- B) False
- C) Error
- D) Null

Answer: B

13. Which method checks if a character is either a letter or a digit?

- A) `char.IsLetterOrDigit()`
- B) `char.IsAlphanumeric()`
- C) `char.IsLetterDigit()`
- D) `char.IsAlphaNum()`

Answer: A

14. What does `char.IsUpper('A')` return?

- A) True
- B) False
- C) Depends on culture
- D) Error

Answer: A

15. Which method determines if a character is a punctuation mark?

- A) `char.IsPunctuation()`
- B) `char.IsSymbol()`
- C) `char.IsSeparator()`
- D) `char.IsMark()`

Answer: A

16. What does `char.IsWhiteSpace('\t')` return?

- A) True
- B) False
- C) Error

- D) Null

Answer: A

17. Which method converts a character to lowercase?

- A) `char.ToLower()`
- B) `char.Lower()`
- C) `char.ConvertToLower()`
- D) `char.DownCase()`

Answer: A

18. What will be the output of `char.ToUpper('m')`?

- A) 'M'
- B) 'm'
- C) Error
- D) 'N'

Answer: A

19. What is returned by `char.ToLower('Z')`?

- A) 'z'
- B) 'Z'
- C) Error
- D) 'A'

Answer: A

String Searching Methods

20. Which method checks if a string contains a substring?

- A) `string.Contains()`
- B) `string.Has()`

- C) `string.Contains()`
- D) `string.Find()`

Answer: A

21. What does `str.StartsWith("Hello")` return if `str = "Hello World"`?

- A) True
- B) False
- C) 0
- D) 1

Answer: A

22. What does `str.EndsWith("World")` return if `str = "Hello World"`?

- A) True
- B) False
- C) -1
- D) 5

Answer: A

23. Which method returns the index of the first occurrence of a substring?

- A) `IndexOf()`
- B) `FirstIndexOf()`
- C) `Search()`
- D) `Find()`

Answer: A

24. What does `"banana".IndexOf('a')` return?

- A) 1
- B) 0
- C) 2
- D) 3

Answer: A

25. What does `"banana".IndexOf('a', 2)` return?

- A) 3
- B) 1
- C) 2
- D) 4

Answer: A

26. What does `"banana".LastIndexOf('a')` return?

- A) 5
- B) 3
- C) 1
- D) 4

Answer: A

String Manipulation Methods

27. What is the purpose of the `Substring()` method?

- A) Extract part of a string
- B) Replace part of a string
- C) Insert into a string
- D) Delete from a string

Answer: A

28. What does `"Hello".Substring(1, 3)` return?

- A) "ell"
- B) "Hel"
- C) "llo"
- D) "lo"

Answer: A

29. Which method inserts a string into another string?

- A) `Insert()`
- B) `Add()`
- C) `Place()`
- D) `Include()`

Answer: A

30. What does `"Hello".Insert(2, "XYZ")` return?

- A) `"HeXYZllo"`
- B) `"XYZHello"`
- C) `"HelloXYZ"`
- D) `"HeXYZlo"`

Answer: A

31. Which method removes characters from a string?

- A) `Remove()`
- B) `Delete()`
- C) `Cut()`
- D) `Erase()`

Answer: A

32. What does `"Hello".Remove(1, 2)` return?

- A) `"Hlo"`
- B) `"Heo"`
- C) `"Hel"`
- D) `"Ho"`

Answer: A

33. What does `" HELLO ".Trim()` return?

- A) "HELLO"
- B) " HELLO"
- C) "HELLO "
- D) " HELLO "

Answer: A

34. Which method removes only leading spaces?

- A) `TrimStart()`
- B) `TrimLeft()`
- C) `TrimBegin()`
- D) `TrimFront()`

Answer: A

35. Which method removes only trailing spaces?

- A) `TrimEnd()`
- B) `TrimRight()`
- C) `TrimBack()`
- D) `TrimFinish()`

Answer: A

36. What does `str.ToUpper()` do if `str = "Hello"`?

- A) Returns "HELLO"
- B) Modifies str to "HELLO"
- C) Returns "hello"
- D) Throws an error

Answer: A

37. What does `str.ToLower()` do if `str = "WORLD"`?

- A) Returns "world"
- B) Modifies `str` to "world"
- C) Returns "WORLD"
- D) Throws an error

Answer: A

String Tokenizing

38. Which method splits a string into tokens?

- A) `Split()`
- B) `Tokenize()`
- C) `Break()`
- D) `Separate()`

Answer: A

39. What does `"a,b,c".Split(',')` return?

- A) `["a", "b", "c"]`
- B) `["a,b,c"]`
- C) `["a", ",", "b", ",", "c"]`
- D) `["a", "b", "c", ""]`

Answer: A

40. Which character is used as a default delimiter when `Split(null)` is called?

- A) White-space characters
- B) Comma
- C) Semicolon
- D) Tab

Answer: A

41. In tokenizing, what is a delimiter?

- A) A character that separates tokens
- B) A character that joins tokens
- C) A type of token
- D) A method name

Answer: A

42. What is the purpose of trimming a string before tokenizing?

- A) To remove unwanted spaces from tokens
- B) To make splitting faster
- C) To change the delimiter
- D) To encrypt the string

Answer: A

Structures

43. What does a structure in C# allow you to do?

- A) Group related variables of different types
- B) Store multiple methods
- C) Create a new class
- D) Define a namespace

Answer: A

44. Which keyword is used to declare a structure?

- A) `struct`
- B) `class`
- C) `record`
- D) `type`

Answer: A

45. In a structure, what are the variables inside called?

- A) Fields
- B) Properties
- C) Members
- D) Elements

Answer: A

46. What access modifier is commonly used for structure fields?

- A) `public`
- B) `private`
- C) `protected`
- D) `internal`

Answer: A

47. Where can a structure be declared?

- A) Inside a namespace
- B) Inside a method
- C) Inside another structure
- D) Both A and C

Answer: D

48. How do you create an instance of a structure?

- A) `Automobile car = new Automobile();`
- B) `Automobile car;`
- C) Both A and B
- D) `new Automobile car;`

Answer: C

49. What does the 'new' operator do when creating a structure?

- A) Initializes fields to default values
- B) Allocates memory only
- C) Calls the constructor
- D) Both A and C

Answer: D

50. How do you access a field of a structure object?

- A) Using the dot operator
- B) Using the arrow operator
- C) Using the index operator
- D) Using the scope operator

Answer: A

51. Can you assign one structure object to another?

- A) Yes
- B) No
- C) Only if they are the same type
- D) Only with explicit casting

Answer: A

52. How are structure objects passed to methods by default?

- A) By value
- B) By reference
- C) By pointer
- D) By name

Answer: A

53. Can you compare two structure objects directly with ==?

- A) No
- B) Yes

- C) Only if overloaded
- D) Only for numeric fields

Answer: A

54. How do you compare two structure objects for equality?

- A) Compare each field individually
- B) Use `Equals()` method
- C) Use `ReferenceEquals()`
- D) Both A and B

Answer: D

55. Can you store structure objects in an array?

- A) Yes
- B) No
- C) Only if they are immutable
- D) Only if they are serializable

Answer: A

56. How do you iterate over an array of structures?

- A) Using a for loop
- B) Using a foreach loop
- C) Both A and B
- D) Neither

Answer: C

57. Can you store structure objects in a List?

- A) Yes
- B) No
- C) Only if they are classes
- D) Only if they implement `IEnumerable`

Answer: A

ImageList and Timer Controls

58. What is the ImageList control used for?

- A) Storing a collection of images
- B) Displaying images in a PictureBox
- C) Both A and B
- D) Editing images

Answer: C

59. What is the maximum image size supported by ImageList?

- A) 256x256 pixels
- B) 512x512 pixels
- C) 1024x1024 pixels
- D) No limit

Answer: A

60. How do you add images to an ImageList control at design time?

- A) Using the Images property in the Properties window
- B) Using code only
- C) Dragging images onto the form
- D) Using the Resources.resx file

Answer: A

61. How do you display an image from an ImageList in a PictureBox?

- A) `pictureBox.Image = imageList.Images[index];`
- B) `pictureBox.Image = imageList[index];`
- C) `pictureBox.Image = imageList.GetImage(index);`
- D) `pictureBox.Image = imageList.Image[index];`

Answer: A

62. How do you get the number of images in an ImageList?

- A) `imageList.Images.Count`
- B) `imageList.Count`
- C) `imageList.ImageCount`
- D) `imageList.Images.Length`

Answer: A

63. What is the Timer control used for?

- A) Executing code at regular intervals
- B) Measuring time between events
- C) Displaying current time
- D) Creating delays

Answer: A

64. In what unit is the Timer control's Interval property measured?

- A) Milliseconds
- B) Seconds
- C) Minutes
- D) Ticks

Answer: A

65. How do you start a Timer?

- A) `timer.Start()`
- B) `timer.Enabled = true`
- C) Both A and B
- D) `timer.Begin()`

Answer: C

66. How do you stop a Timer?

- A) `timer.Stop()`
- B) `timer.Enabled = false`
- C) Both A and B
- D) `timer.End()`

Answer: C

67. Which event is raised by the Timer control?

- A) Tick
- B) Elapsed
- C) Interval
- D) TimeChanged

Answer: A

68. In the slideshow example, how is the next image displayed?

- A) Increment index and assign image to PictureBox
- B) Use Random class
- C) Use Timer only
- D) Use `ImageList.Next()`

Answer: A

69. What is used in the Random Card example to pick a random image?

- A) Random class
- B) Timer class
- C) Shuffle method
- D) `Array.Sort()`

Answer: A

70. How do you ensure images in an ImageList are displayed correctly?

- A) Set ImageSize property to match image dimensions
- B) Use only .bmp files
- C) Use only small images
- D) Set ColorDepth to HighColor

Answer: A

Additional Character Testing

71. Which of the following is true about the `char.IsDigit()` method?

- A) It returns true for '0' through '9'
- B) It returns true for any numeric character in any culture
- C) It works only with strings
- D) It returns false for '0'

Answer: A

72. What does `char.IsLetter('@')` return?

- A) False
- B) True
- C) Error
- D) Null

Answer: A

73. Which method checks if a character is a white-space character?

- A) `char.IsWhiteSpace()`
- B) `char.IsSpace()`
- C) `char.IsBlank()`
- D) `char.IsEmpty()`

Answer: A

74. What is the result of `char.ToLower('A')`?

- A) 'a'
- B) 'A'
- C) '1'
- D) Error

Answer: A

75. What is the result of `char.ToUpper('z')`?

- A) 'Z'
- B) 'z'
- C) 'Y'
- D) Error

Answer: A

Additional String Methods

76. Which method would you use to check if a password contains an '@' symbol?

- A) `Contains()`
- B) `Has()`
- C) `Includes()`
- D) `Find()`

Answer: A

77. What does `"Hello".Contains("ell")` return?

- A) True
- B) False
- C) 1
- D) 2

Answer: A

78. What does `"Hello".StartsWith("He")` return?

- A) True
- B) False
- C) 0
- D) 2

Answer: A

79. What does `"Hello".EndsWith("lo")` return?

- A) True
- B) False
- C) 3
- D) 5

Answer: A

80. What does `"Hello".IndexOf('e')` return?

- A) 1
- B) 0
- C) -1
- D) 2

Answer: A

81. What does `"Hello".IndexOf('l', 3)` return?

- A) 3
- B) 2
- C) -1
- D) 4

Answer: A

82. What does `"Hello".LastIndexOf('l')` return?

- A) 3
- B) 2
- C) 1
- D) 4

Answer: A

83. What does "Hello".Substring(2) return?

- A) "llo"
- B) "Hel"
- C) "lo"
- D) "ell"

Answer: A

84. What does "Hello".Substring(1, 3) return?

- A) "ell"
- B) "Hel"
- C) "llo"
- D) "lo"

Answer: A

85. What does "Hello".Insert(2, "XYZ") return?

- A) "HeXYZllo"
- B) "HelXYZlo"
- C) "XYZHello"
- D) "HelloXYZ"

Answer: A

86. What does "Hello".Remove(2, 2) return?

- A) "Heo"
- B) "Hlo"

- C) "Hel"
- D) "He"

Answer: A

87. What does " Hello ".Trim() return?

- A) "Hello"
- B) " Hello"
- C) "Hello "
- D) " He llo "

Answer: A

88. What does " Hello ".TrimStart() return?

- A) "Hello "
- B) " Hello"
- C) "Hello"
- D) " He llo "

Answer: A

89. What does " Hello ".TrimEnd() return?

- A) " Hello"
- B) "Hello "
- C) "Hello"
- D) " He llo "

Answer: A

90. What does "Hello".ToUpper() return?

- A) "HELLO"
- B) "hello"
- C) "Hello"
- D) "HeLLo"

Answer: A

91. What does `"HELLO".ToLower()` return?

- A) "hello"
- B) "HELLO"
- C) "Hello"
- D) "hELLO"

Answer: A

92. What does `"a,b,c".Split(',')` return?

- A) ["a", "b", "c"]
- B) ["a,b,c"]
- C) ["a", ",", "b", ",", "c"]
- D) ["a", "b"]

Answer: A

93. What does `"one two three".Split(null)` return?

- A) ["one", "two", "three"]
- B) ["one two three"]
- C) ["one", " ", "two", " ", "three"]
- D) ["one", "two"]

Answer: A

94. What is the purpose of the Trim method before tokenizing?

- A) Remove leading/trailing spaces from tokens
- B) Make the string shorter
- C) Change the delimiter
- D) Encrypt the string

Answer: A

Additional Structure Questions

95. In a structure, can fields be of different data types?

- A) Yes
- B) No
- C) Only if they are numeric
- D) Only if they are reference types

Answer: A

96. How do you declare a structure named Person with fields name and age?

- A)

```
struct Person
{
    public string name;
    public int age;
}
```

- B)

```
class Person
{
    public string name;
    public int age;
}
```

- C)

```
record Person(string name, int age);
```

- D)

```
type Person { string name; int age; }
```

Answer: A

97. How do you create an instance of the Person structure?

- A) `Person p = new Person();`

- B) `Person p;`
- C) Both A and B
- D) `new Person p;`

Answer: C

98. How do you set the name field of a Person object?

- A) `p.name = "John";`
- B) `p->name = "John";`
- C) `Person.name = "John";`
- D) `name(p) = "John";`

Answer: A

99. Can you pass a structure to a method by reference?

- A) Yes, using the `ref` keyword
- B) No, structures are always passed by value
- C) Yes, using the `out` keyword
- D) Both A and C

Answer: D

100. How do you compare two Person objects for equality?

- A) Compare each field individually
- B) Use `p1 == p2`
- C) Use `p1.Equals(p2)`
- D) Both A and C

Answer: D

101. Can you store Person objects in a List?

- A) Yes
- B) No
- C) Only if Person is a class

- D) Only if Person implements IComparable

Answer: A

102. How do you add a Person object to a List?

- A) `list.Add(person);`
- B) `list.Append(person);`
- C) `list.Insert(person);`
- D) `list.Push(person);`

Answer: A

Additional ImageList and Timer Questions

103. What is the purpose of the ImageList control?

- A) To store multiple images for use by other controls
- B) To display a single image
- C) To edit images
- D) To convert image formats

Answer: A

104. What property sets the size of images in an ImageList?

- A) `ImageSize`
- B) `Size`
- C) `ImageDimensions`
- D) `PictureSize`

Answer: A

105. How do you display the first image from an ImageList in a PictureBox?

- A) `pictureBox.Image = imageList.Images[0];`
- B) `pictureBox.Image = imageList[0];`
- C) `pictureBox.Image = imageList.GetImage(0);`

- D) `pictureBox.Image = imageList.ImageAt(0);`

Answer: A

106. How do you get a random image from an ImageList?

- A) Use Random class to generate a valid index
- B) Use `imageList.Images.Random()`
- C) Use `imageList.GetRandomImage()`
- D) Use Shuffle method on Images collection

Answer: A

107. What does the Timer control's Interval property represent?

- A) Time in milliseconds between Tick events
- B) Total duration before Timer stops
- C) Number of Tick events to occur
- D) Delay before the first Tick event

Answer: A

108. How do you handle the Timer's Tick event?

- A) Write code in the timer_Tick event handler
- B) Override the OnTick method
- C) Use a delegate
- D) Both A and C

Answer: A

109. In a slideshow, how do you move to the next image automatically?

- A) Use a Timer to change the image at intervals
- B) Use a Button click event only
- C) Use a Thread to sleep and change image
- D) Use a BackgroundWorker

Answer: A

110. How do you stop an automatic slideshow?

- A) `timer.Stop();`
- B) `timer.Enabled = false;`
- C) Both A and B
- D) `timer.Pause();`

Answer: C

Edge Cases and Special Scenarios

111. What is the result of `"Hello".Contains('e')`?

- A) True
- B) False
- C) Error
- D) 1

Answer: A

112. What is the result of `"Hello".Contains("E")`?

- A) False
- B) True
- C) 0
- D) Error

Answer: A

113. What does `char.IsPunctuation('!')` return?

- A) True
- B) False
- C) Error
- D) Null

Answer: A

114. What does `char.IsPunctuation('a')` return?

- A) False
- B) True
- C) Error
- D) Null

Answer: A

115. What does `char.IsWhiteSpace(' ')` return?

- A) True
- B) False
- C) Error
- D) Null

Answer: A

116. What does `char.IsWhiteSpace('A')` return?

- A) False
- B) True
- C) Error
- D) Null

Answer: A

117. What is the result of `"Hello".IndexOf("lo")`?

- A) 3
- B) 2
- C) 4
- D) -1

Answer: A

118. What is the result of `"Hello".IndexOf("xyz")`?

- A) -1
- B) 0
- C) 1
- D) Error

Answer: A

119. What is the result of `"Hello".LastIndexOf("l")`?

- A) 3
- B) 2
- C) 1
- D) -1

Answer: A

120. What is the result of `"Hello".LastIndexOf("x")`?

- A) -1
- B) 0
- C) 1
- D) Error

Answer: A

121. What is the result of `"Hello".Substring(5)`?

- A) Throws `ArgumentOutOfRangeException`
- B) Returns empty string
- C) Returns "Hello"
- D) Returns null

Answer: A

122. What is the result of `"Hello".Substring(1, 10)`?

- A) Throws `ArgumentOutOfRangeException`
- B) Returns "ello"
- C) Returns "Hello"
- D) Returns null

Answer: A

123. What is the result of `"Hello".Insert(5, " World")`?

- A) "Hello World"
- B) "Hello"
- C) Throws `ArgumentOutOfRangeException`
- D) " WorldHello"

Answer: A

124. What is the result of `"Hello".Remove(5)`?

- A) "Hello"
- B) Throws `ArgumentOutOfRangeException`
- C) ""
- D) "Hell"

Answer: A

125. What is the result of `"".Trim()`?

- A) ""
- B) " "
- C) null
- D) Throws exception

Answer: A

126. What is the result of `null.ToUpper()`?

- A) Throws NullPointerException
- B) Returns null
- C) Returns ""
- D) Returns "NULL"

Answer: A

127. What is the result of `"HELLO".ToLower()` ?

- A) "hello"
- B) "HELLO"
- C) "Hello"
- D) "hELLO"

Answer: A

128. What is the result of `"a,b,c".Split(',') [1]` ?

- A) "b"
- B) "a"
- C) "c"
- D) ","

Answer: A

129. What is the result of `"one two three".Split(null) [0]` ?

- A) "one"
- B) "two"
- C) "three"
- D) " "

Answer: A

130. In the CSV example, what delimiter is used to split each line?

- A) Comma (',')
- B) Semicolon (';')

- C) Colon (':')
- D) Pipe ('|')

Answer: A

Wish you all the best!
Fathy.